

Getting Acquainted

C-200MT Software for GERBERcutters



The C-200MT software is designed to operate on various GERBERcutters in various work environments. C-200MT performs interface and cutter control functions on all new GERBERcutters.

Multitasking

The C-200MT program version 1.6.x or greater runs only in Windows NT 4. You may already be familiar with Windows. If not, refer to the *Windows NT Quick Reference Guide*. Your company will appreciate the fact that Windows NT allows multitasking. This means additional tasks can be performed while a cut file/marker is being cut. These tasks can include:

- Viewing a different cut file
- Creating a job queue
- Viewing system setups
- Using File Manager
- Generating reports

Supported by C-200MT

GERBERcutters

The C-200MT program performs user interface and cutter control functions on the following GERBERcutters:

- GT 3250/5250/7250
- GTxL
- S-3200/5200/7200
- S-91 TCW, ZVC, Static

The cutter type is selected when installing the software for the first time. Your GERBERcutter was shipped from the factory with the software pre-installed.

File Formats

Currently, your GERBERcutter can process the following file types: AAMA data, AccuMark marker data (version 7.2 or greater) and GERBERcutter data (274-D).

GERBERcutter Data Formats

The C-200MT program supports the following standard cut data formats: 3.3 English, 4.2 English, and 5.1 Metric.

Native Languages

The C-200MT program supports native languages, including:

- All 8-bit character sets supported by Windows NT 4
- Those 16-bit languages for which a Windows NT 4 version exists (example: Japanese Windows NT)

Networks

The C-200MT program supports connectivity to other systems on a Local Area Network. All networks capable of running with Windows NT 4 are supported.

Accessing the Program

The C-200MT program is the first C-200 cutter controller PC software that runs in Microsoft Windows NT. Your company will appreciate the fact that Windows NT allows multitasking. This means that additional utilities or options, such as PreProcessor or CutView can be used while a cut file/marker is being cut.

To start up the C-200MT program

1. Turn on the main power circuit breaker for the cutter. Refer to the GERBERcutter operator's manual or Getting Started.
2. Press **Enter** to select Windows NT Workstation.
3. When prompted, press **Ctrl + Alt + Delete** to log on.
4. When prompted for a password, type your password and then press **Enter** or click **OK** to display the Main Cutter Control screen.

Note: The system is shipped from the factory without an NT password. Press Enter to continue.

5. Double-click the C-200MT icon on the Windows desktop.

Closing the Program

This section gives the steps necessary to close the C-200MT program and shut down the cutter.

CAUTION: To avoid losing data, always quit the C-200MT program before shutting down Windows NT. Also, shutdown Windows NT before turning off the main circuit breaker for the cutter.

To quit the C-200MT program

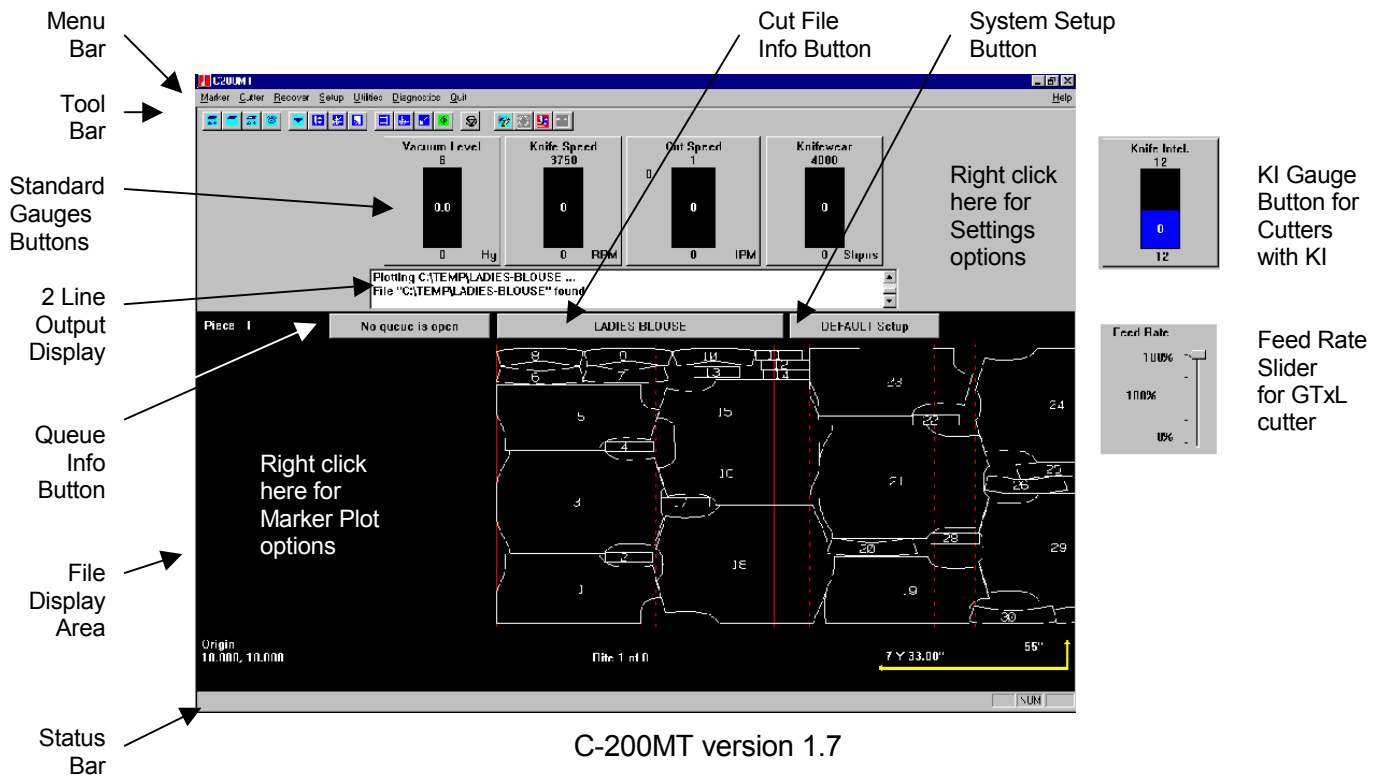
1. Make sure that the cutter is idle (not processing data) and that the servo power is disabled.
2. From the Quit menu, select **Confirm**.

To shut down the cutter

1. Click on the Start button in the Task bar. Do one of the following:
 - Select **Shutdown**.
 - Press ALT + F4.
2. In the Shutdown dialog box, click the radio button next to Shutdown. Then press **Enter** or click the **OK** button.
3. When prompted that it is safe to turn off your computer, turn off the main power circuit breaker.

C-200MT Main Cutter Control Screen

When you turn on your GERBERcutter, the following screen displays. Features in this screen are explained in the following sections.



Note: If your cutter is running C-200MT version 1.5.x or earlier, refer to the Getting Acquainted topic in the C-200MT online help on your system.

Menu Bar

The menu bar on the Main Cutter Control screen has drop-down menus that allow access to the interface and cutter control functions in the C-200MT program. Each menu choice is called a command.

Menus available in the Main Cutter Control screen:

Menu Item	Use commands to
File	Open, close, rerun, and manage marker files.
Cutter	Operate the cutter.
Recover	Recover from an EStop or system power down.
Setup	Configure settings that determine how the cutter operates; change passwords.
Utilities	View markers; run a job queue; Preprocess markers, Options: automatically clean the table surface; create CIDB reports.
Diagnostics	Check cutter operation.
Help	Find help on how to use features in the C-200MT program and review information about the current software release.
Quit	Exit the C-200MT program

Tool Bar

The tool bar on the Main Cutter Control screen provides an additional way to perform frequently-used tasks for operating the cutter. Click on an icon with the pointing device to select.

Hold the cursor over a tool bar icon to view the name and corresponding function key you can press on the keyboard, if preferred.

Icon Keys and BCP Buttons

Notice that icons on several tool bar buttons correspond to those on switches on the beam control panel at the cutter. For GERBERcutters with colored icon keys on the keyboard, these also are identical to icons on the tool bar buttons.

<u>Icon</u>	<u>Command</u>	<u>Function Key</u>	<u>Use To</u>
	Open Cutfile	F1	Select cut file or AccuMark marker, and open it for processing.
	Piece Search	F2	Find and cut a specific piece in a marker.

<u>Icon</u>	<u>Command</u>	<u>Function Key</u>	<u>Use To</u>
	Restore State	F3	Recover or restore state after a power loss or EStop condition.
	View Cutfile	F4	View cut data including dry haul path, bite lines, drill points, and data points. The CutView utility also allows you to zoom in/out, select individual bites and pieces within bites, animate the cutting process, and check piece data.
	Setup	F5	Access Settings in order to change setups. The C-200MT program provides default setups for running a cutter. Setups are suitable for a wide variety of applications.
	Knife Speed	F6	Adjust knife speed for the current session or set as defaults in the setup.
	Move To Origin	F7	Move the cutter head to the last established origin.
	Sharpen	F8	Sharpen the knife when the cutter is idle.
	Vacuum	F9	Turn table vacuum ON or OFF.
	Origin	F10	Set the "start" position (X0,Y0) of a marker on the table.
	Initialize	F11	Initialize, or identify, the zero position for the table in X and Y (X0,Y0). This is also called the home.
	Start Cutting	F12	Begin the cutting process.
	Stop Cutting	Pause	Stop the cutting process.
	Queue Manager		Create job queues that allow the cutter to automatically process multiple.
	AccuMatch		Access the AccuMatch option, if available.
	Preprocessor		Access the Preprocessor to optimize cut data files.
	CIDB Report Generator		Access the report generator, if CIDB option is available.

Cut File Information Button

This button on the Main Cutter Control screen displays the cut file name when one is open. Click the button to display the Cut File Information dialog box. This dialog box shows the cut file name and path as well as the length, width, number of bites, and number of pieces in the open cut file. The dialog box also allows you to open/close a cut file/marker or re-run a file.

Information concerning the current file and job displays in this window. This can include:

- Name of setup used
- Cut path (drive and directory)
- Cut File/Marker name
- Length and width of marker
- Bite number currently being processed
- Piece number currently being processed
- Origin in X and Y
- Job queue, if any
- Cutter status mode (determines which commands can be used currently)

Queue Information Button

This button on the Main Cutter Control screen displays the Queue name when one is open. Click the button to display the Queue Information dialog box. This dialog box allows you to:

- Check the current job, if any
- Open a queue
- Close the current queue
- Select the next job in the current queue
- Open Queue Manager to edit the current or create a new queue

Click **Done** to close this dialog box.

System Setup Button

This button on the Main Cutter Control screen displays the system setup dialog box. From here, you can load a new setup or save a setup you created. Clicking this button is the same as selecting Settings from the Setup menu.

Two Line Output Display

This display on the Main Cutter Control screen relays informational messages that do not require any user acknowledgement. You can scroll through the messages using the scroll buttons on the window.

These can include:

Initializing...
Command Executed Move Complete

Settings Options

To access Settings Options (explained below), right click anywhere in the gray bar area, except on a gauge or other button.

To turn an option on, click that option. A button is created and displays as long as an option is active.

To turn an option off, click on the button displayed for that option.

Option	Use To
Turn On Dry Run	Check the cut file/marker by processing without cutting.
Enable Single Stepping	Process cut file data one block at a time.
Enable Continuous Run	Reprocess an open cut file until you manually stop the process.
Setup Data Scaling	Change scaling to adjust the size of a marker to the size of the available material.
Setup a Fabric Alignment Angle	Set a different angle before cutting a marker.
Setup a Clipping Window	Prevent the knife from cutting outside of one or more edges of the spread.

Status Bar

When a menu command is highlighted, a brief description about the command displays in the Status bar at the bottom of the Main Cutter Control screen.

Gauges on Main Screen

Status gauges appear on the Main Cutter Control screen as soon as servo power is enabled. They replace the Enable Servo Power prompt. These gauges frequently-updated indicators that show:

- Vacuum Level
- Knife Speed
- Cut Speed
- Knifewear
- Knife Intel.
- Feed Rate Slider

The Percent Complete, although not among the group of gauges described above, indicates cut file completion.

Knife Speed Gauge

Shows current knife reciprocating speed in revolutions per minute (rpm).

Click the gauge button to access the Knife Speeds dialog box.

Cut Speed Gauge

Shows current cutting speed in inches or centimeters per minute for the cutting head. The Cut Speed gauge indicates the slowdown by a decrease in the width of the green bar indicator.

Click the gauge button to access the Cut Speed dialog box.

Knife Wear Gauge

Shows the number of sharpens that have occurred since the last knife; it also indicates how many sharpens remain before the knife needs to be changed. When the maximum number of sharpens has been reached, the bar graph turns red and the Knife Life Exceeded prompt displays.

Click the Knife Wear gauge button to access the Knife Wear and Sharpening dialog box.

Knife Intelligence Gauge

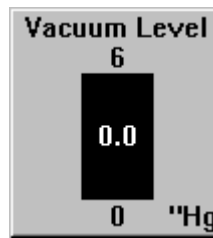
Shows the current change in knife heading to compensate for lateral forces on the knife; angle change is determined by Knife Intelligence (KI). The Knife Intel. gauge only displays on cutters that have Knife Intelligence.

Click the gauge button to access the Knife Intelligence dialog box.

Note: If you temporarily disable Knife Intelligence, the gauge will display a red bar by default.

Vacuum Level Gauge

This button on the Main Cutter Control screen shows current vacuum level in inches or centimeters per mercury. Click the gauge button to access the Vacuum Settings dialog box.



When the level is too low the bar graph turns red. The message, **Vacuum Low Caution**, displays when the vacuum is too low for processing the current marker.

Percent Complete Gauge

This button on the Main Cutter Control screen displays at the bottom of the C-200MT plot window area. It shows the percentage of cutting completed so far for the current marker; the percentage is based on distance cut relative to total cut distance.

Feed Rate Slider

On a GTxL GERBERcutter, this slider allows you to view and change the current feed rate.

File Display Area

When a cut file/marker is opened, this portion of the Main Cutter Control screen shows the current bite and indicates the piece currently being cut. The status of the plot is updated several times a second. Information concerning the current file and job displays in this window. This can include:

- Piece number currently being processed
- Cutter status mode
- Origin in X and Y
- Bite number currently being processed
- Job queue, if any
- Length and width of marker

File Display Options

To access File Display Options (explained below), right click in the File Display area. Right clicking in the screen will provide the following options, depending on whether a cut file/marker is open or not:

With Cut File/Marker Open	
Option	Use To
View	Access the CutView utility.
Close	Close currently displayed cut file/marker.
About	Display the Cut File Information dialog box. This dialog box shows the cut file/marker name and path as well as the length, width, number of bites, and number of pieces in the open cut file. The dialog box also allows you to open/close a cut file/marker or re-run a file.
Rerun	Reopen and display the last cut file/marker opened. The system will display cut file/marker's name in the menu. Note: If you attempt to rerun a currently displayed cut file/marker, the system will prompt you to close and reopen that file.
Open a Different Cutfile	Open a different cut file/marker than the one currently open.
With Cut File/Marker Closed	
Option	Use To
Rerun	Reopen and display the last cut file/marker opened. The system will display the cut file/marker's name in the menu. Note: If you attempt to rerun a currently displayed cut file/marker, the system will prompt you to close and reopen that file.
Open a Cut file	Open a cut file/marker.

Piece Colors in File Display Area

In the File Display area, piece color changes to differentiate those that have not yet been cut.

1. Pieces are first white.
2. When a piece is about to be cut, it turns light blue.
3. As the knife cuts a piece, the cut line becomes yellow.
4. After a piece is cut, the piece number turns yellow, too.

Options for GERBERcutter Systems

Gerber provides options to enhance the performance of your GERBERcutter. Here is a list of hardware and software available. For more information, call your local Gerber Sales and Service office.

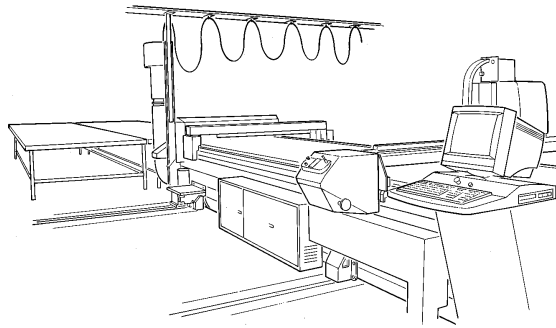
- Lateral Drive
- Right Side Option (RSO)
- CTOT
- Barcode Reader
- Drill
- Bristle Cleaner
- Resealer
- Power Processor GPP-200
- AccuMatch
- Automatic Tube Centering
- Y Edge Alignment

Note: Not all options are available on all GERBERcutters.

Right Side Option

The Right Side Option (RSO) allows you run the GERBERcutter from the right side of the cutting table. This lets you spread material with its straight edge on the right side, which is more practical for some applications. On a right-side option GERBERcutter, both the operator console and beam control panel are located on the right side of the cutting table as seen from the take-on end.

An RSO option is available for all GERBERcutter systems. GTxL GERBERcutter systems can be reconfigured on-site for an RSO option. Shown below is an S-91 TCW system with the RSO option.



Conveyorized Take-off Table (CTOT)

Depending on the GERBERcutter model, a CTOT is built into the cutter or available as a moveable unit. Cut pieces and scrap material moves off of the cutting table onto the CTOT. If the CTOT is placed in front of another table, pieces can continue to move onto that table for picking and bundling. Otherwise, cut pieces are picked up directly from the CTOT and scrap material can drop away into a bin or onto the floor.

Lateral Drive Option

A lateral drive option is available for most GERBERcutter systems. This option allows one cutter to move from spreading table to spreading table. Included in the option are:

- Platform wheels which ride in tracks in the cutting room floor
- Lateral drive controls
- Vacuum exhaust stack on mast
- Mast and overhead trolley to carry electric cabling

Resealer Option

The resealer uses plastic film to automatically cover, or seal, the cut area of the lay. Covering areas on the table where pieces have already been cut, improves cut quality for remaining pieces by maintaining vacuum. Resealers on higher ply GERBERcutter systems have a bar that can be raised to allow uncompressed pieces and material to be conveyed off of the table.

Bristle Cleaner Options

Removing lint and debris from the table surface promotes vacuum flow through the BRISTLE SQUARES. Options available for cleaning the work surface depend on the GERBERcutter model.

- Detachable vacuum hose
- Automatic Bristle Cleaning option (See Routine Maintenance chapter)

Bar Code Reader

This option, available for all GERBERcutter systems, allows you to use a bar code reader to open individual cut files or job queues. The Bar Code Reader is capable of scanning cut file names 20 characters in length. This was tested using Code 39 bar codes.

To turn on this option

1. Open the Parameter Manager.
2. The Bar Code Reader Option parameter in the Options and Modes category must be set to ON.
3. To open job queues with the Bar Code Reader, set the Read Queue Name Bar Codes parameter in the Queue Manager category to YES.

When this feature is turned on, all bar codes read (except setup files *.CNF) are assumed to be queues and are automatically opened for running. If this feature is turned off, each bar code read will be opened as single cut file.

POWER Processor GPP-200

The POWER Processor™ lets you change cut files so that the GERBERcutter works more efficiently. Input to the POWER Processor is a cut file produced on a marking and grading system such as Gerber's AccuMark or MicroMark. Output is a modified cut file. Altering a cut file can improve cut quality, cut speed, material use, and cut file efficiency. The POWER Processor lets you:

- Change piece cutting sequence
- Change the start point of a piece
- Add or remove knife lift and plunge points
- Add or remove Opstop points for plaid/stripe matching

AccuMatch



With the Gerber AccuMatch system, you can check match points and make any necessary adjustments with accuracy and speed. A color video CCD module, or digital camera, captures images of the spread at all matching positions (Opstops) in a cut file and displays on-screen an image of the material at this location. You can accept the match point as shown or reposition a cross hair cursor relative to the fabric design.

The AccuMatch system option runs on S-3200 and GT 3250 GERBERcutters and includes the following:

Hardware

- Camera with adjustable lens
- Light
- Mounting fixture
- Camera head cover
- Cables
- High performance video board
- Auxiliary slew control (joystick) at operator console

Software

Check with your Gerber Sales representative for the appropriate AccuMatch and C-200MT software versions for your cutter.

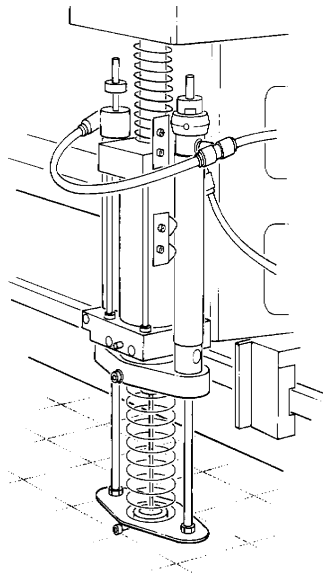
Drill Options

All GERBERcutter systems support a drill option. The following options are available, depending on the cutter model. Gerber also provides drill bits in a wide variety of sizes and types.

Drill options available:

- Electric
- Pneumatic/Electric

An example of an electric drill is shown.



Drill Options — Electric

Cutter	Drill 1	Drill 2
GTxL, GT 7250, S-7200, S-91	Electric	Electric
GT 5250, S-5200	Electric	Ask about availability

Drill Options — Pneumatic/Electric

Cutter	Drill 1	Drill 2
GT 3250, S-3200	Pneumatic	Electric
	Electric (heavy duty)	

Centering and Alignment Options

Y Edge Alignment Option

The Y-edge Alignment option uses a roller assembly to track the location of the edge of a spread along the Y-axis of the cutting table and relay this information by pneumatics to the cutter. This allows the system to accurately position and cut pieces.

For information about this option, which is available for S-91 ZVC cutters only, contact your Service Representative.

Automatic Tube Centering Option

The Automatic Tube Centering option can be used to find, accurately and automatically, the centerline origin of a spread. Centerline Origin is a C-200MT feature used for cutting tubular goods or any other pattern that must be centered on a spread.

Features for Automatic Tube Centering include:

- Automatic determination of the center of the spread.
- Automatic fabric alignment.
- Automatic scaling of the Y-axis.
- Adjustable edge sensor width.
- Edge sensor diagnostic for setting up the option.

This option is available for GT 5250/7250 and S-5200/7200 cutters except those with the RSO or Drill options. For more information, contact your Gerber Service Representative.

